

Project Report: Cryptographic Algorithms Implementation

By: Vincent Fealy
COSC 472

This project implements four cryptographic algorithms, each serving a unique purpose in the encryption and decryption of text data and hashing. The algorithms included are the Caesar Cipher, Monoalphabetic Cipher, Polyalphabetic Cipher, and the Toy Tetragraph Hash (TTH). Each section of this report provides an overview of each algorithm, potential weaknesses, and possible solutions.

Description of Algorithms

The Caesar Cipher is a substitution cipher where each letter in the plaintext is replaced by a letter a fixed number of positions down the alphabet. For example, with a shift of 3, 'A' becomes 'D', 'B' becomes 'E' and so on. This algorithm is simple and effective for basic encryption but offers minimal security.

The Monoalphabetic Cipher uses a fixed substitution method based on a keyword. The keyword is written first, removing duplicate letters, followed by the remaining letters of the alphabet. For instance, the key "zebras" results in a mapping that substitutes letters based on the provided key, making it less predictable than the Caesar Cipher.

The Polyalphabetic Cipher uses multiple Caesar Ciphers based on the characters in a key. Each letter of the key determines the shift for the corresponding letter in the plaintext, creating a more complex encryption. For example, with the key "ad", the first letter might be shifted by 0 (for 'a'), and the second by 3 (for 'd'), effectively using different substitution ciphers throughout the text.

The Toy Tetragraph Hash is a simplified hash function that processes input text by breaking it into segments of four characters and applying basic mathematical operations to generate a hash value. This approach is less complex than standard hash functions like SHA but provides an elementary demonstration of hashing concepts.

Discussion of Algorithms

The Caesar Cipher presents several weaknesses that can compromise its security. One significant vulnerability is its susceptibility to frequency analysis; as a simple substitution cipher, attackers can analyze the frequency of letters in the ciphertext to deduce the key. Additionally, the cipher has a limited key space, with only 25 possible shifts, making it relatively easy to break through brute-force attacks. To enhance security, one possible solution is to use a more complex key or incorporate additional layers of encryption, such as combining the Caesar Cipher with another cipher.

The Monoalphabetic Cipher also shares weaknesses similar to the Caesar Cipher, primarily its susceptibility to frequency analysis. Statistical analysis of letter frequency can reveal the substitution mapping that allows attackers to break the encryption. Also once the fixed substitution mapping is known, it can be exploited easily. To protect these vulnerabilities, putting in a dynamic key generation method could be helpful. This approach would allow for frequent changes to the substitution mapping, or alternatively, switching to a more robust cipher could provide greater security.

The Polyalphabetic Cipher has its own challenges, especially regarding key length sensitivity. If the key is shorter than the plaintext, the key will repeat and that makes it vulnerable to analysis. The complexity of managing and generating keys securely can cause difficulties. To fix these issues, using longer and even more randomized keys can minimize patterns in the ciphertext, this would reduce vulnerability to analysis. Employing key stretching techniques can enhance secure key management practices as well.

The Toy Tetragraph Hash (TTH) has many weaknesses as well. One of its main vulnerabilities is collision susceptibility, due to its simplistic design, which may lead to the same hash being produced for different inputs. The TTH lacks the necessary complexity to be used for serious cryptographic applications, making it susceptible to preimage attacks. A preimage attack occurs when an attacker attempts to find an input that corresponds to a specific hash output, which is easier with simpler hashing algorithms like TTH. To prevent this a recommended solution is to utilize a more sophisticated hash function, like SHA-256. SHA-256 (Secure Hash Algorithm 256-bit) is widely used for cryptographic security. It produces a fixed 256 bit hash value from input data that makes it highly resistant to both collision and preimage attacks. By using SHA-256, the hashing process becomes significantly more reliable and secure for cryptographic applications.

References

1. Menezes, Alfred J., et al. *Handbook of Applied Cryptography*. CRC Press, 1997.
2. Katz, Jonathan, and Yehuda Lindell. *Introduction to Modern Cryptography: Principles and Protocols*. CRC Press, 2020.
3. Rivest, Ronald L. "The MD5 Message-Digest Algorithm." *RFC 1321*, 1992.
4. "What are preimage and collision resistance in hashing?" *Cryptography Stack Exchange*, 2014, <https://crypto.stackexchange.com/questions/29096/what-are-preimage-and-collision-resistance-in-hashing>.
5. Stinson, Douglas R. *Cryptography: Theory and Practice*. Chapman and Hall/CRC, 2006.
6. Ferguson, Niels, Bruce Schneier, and Tadayoshi Kohno. *Cryptography and Network Security: Principles and Practice*. 3rd ed., Pearson, 2010.